

Satya Mohan Samanthakurthi

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SUMMARY

Azure Data Engineer with **3+ years** of Cognizant experience building cloud data pipelines, Azure Databricks workflows, PySpark transformations, Azure SQL datasets, BI reporting layers, and Microsoft Fabric learning projects across healthcare and finance domains. Migrated **50+ SQL scripts**, rebuilt **10+ legacy SSIS/SSRS workflows**, maintained **120+ SSIS workloads**, reconciled **10M+ records**, and improved report performance by up to **30%**. Hands-on with **Azure Databricks, PySpark, Spark SQL, Azure Data Factory, Delta Lake, Medallion Architecture, Microsoft Fabric, OneLake, Lakehouse, RPA, HIPAA-aware data handling, and GDPR principles**. Worked with data analysts, testers, SMEs, and team leads on R&D, source-to-target mapping, migration POCs, data validation, and production release checks.

PROFESSIONAL EXPERIENCE

Cognizant, Azure Data Engineer / Associate Engineer, Hyderabad, India Jul 2024 – Dec 2024

Client: Optum Finance / UnitedHealth Group

- Built Azure Databricks pipelines using Apache Spark, PySpark, Spark SQL, Azure SQL, and Databricks Jobs to ingest, transform, and validate finance datasets for **5+ business units**.
- Migrated **50+ SQL scripts** into configuration-driven Databricks notebooks while preserving business rules, joins, filters, reconciliation logic, and report output structures.
- Designed PySpark transformations covering joins, aggregations, deduplication, schema alignment, NULL handling, audit columns, and automated row-count checks across **20+ datasets**.
- Improved pipeline execution performance by **25%** through Spark SQL refactoring, partition-aware transformations, reduced repeated logic, and notebook parameterisation.
- Built Power BI dashboards and DAX measures on curated finance datasets, enabling stakeholders to monitor **10+ KPIs** and reduce manual Excel reporting by approximately **30%**.
- Worked with data analysts, testers, SMEs, and team leads to complete R&D, field mapping, transformation validation, defect triage, and release readiness checks.
- Promoted code and report changes across multiple environments using Git/TFS, CI/CD controls, and validation evidence.

Cognizant, Cloud Migration Engineer / Associate Engineer, Hyderabad, India Oct 2023 – Jun 2024

Client: CareSource – BI Reporting 2.0

- Delivered a **2-month** cloud migration POC from on-premises SSIS/SSRS to Azure Databricks and Power BI, improving report performance by **30%** and securing client sign-off.
- Re-engineered legacy SSIS ETL logic into modular Databricks SQL and PySpark ELT pipelines with parameterised notebooks, reusable transformations, and workflow-level scheduling.
- Designed a Lakehouse-style migration layer using Delta Lake concepts, controlled table structures, audit columns, row-count checks, and report validation datasets.
- Reconciled **10M+ records** between SQL Server, SSIS outputs, and Databricks datasets using row counts, aggregate comparisons, exception reports, and source-to-target rules.
- Rebuilt **10+ SSIS/SSRS workflows** as cloud-native Databricks pipelines while maintaining HIPAA-aware data handling and preparing datasets for future ML feature engineering.
- Collaborated with data analysts, testers, SMEs, and team leads to compare legacy outputs, validate KPI rules, document mapping decisions, close UAT defects, and prepare sign-off packs.

Cognizant, Programmer Analyst – Data Engineering & BI, Hyderabad, India Nov 2021 – Sep 2023

Client: CareSource – Reporting

- Developed **200+ T-SQL stored procedures, views, functions, indexed views, triggers, and constraints** to support healthcare finance, claims, provider, and member reporting for **5+ business units**.
- Built and supported **120+ SSIS workloads** integrating SQL Server, flat files, Excel, ODBC, and OLE DB sources into reporting-ready datasets.
- Designed **30+ SSRS paginated reports** with drill-down views, matrix layouts, grouped tablix regions, parameterised filters, and state-specific reporting logic.
- Improved average query performance by **20%** through indexing strategies, execution-plan review, join optimisation, stored procedure refactoring, and filter tuning.
- Automated bulk data workflows using SSIS control-flow and data-flow components including Execute SQL Task, Foreach Loop, Lookup, Merge Join, Conditional Split, Derived Column, and Bulk Insert.
- Supported **200+ report and ETL releases** across DEV, INT, CRT, UAT, EUAT, PRE-PROD and PROD environments with **zero P1 incidents** across a **2-year** tenure.

EDUCATION & ACADEMIC PROJECTS

MSc, Artificial Intelligence for Business — National College of Ireland, Dublin

Jan 2025 – Jan 2026

- Developed a Hybrid Ant Colony Optimisation model for last-mile delivery using the LaDe dataset with **1.4M+ delivery trajectories**; KNN-ACO achieved a **2.46 km** best route, **73.85%** improvement over random initialisation, and **68.46%** Hit Rate @ Top 3 across **30** test routes.
- Designed a UiPath RPA proof-of-concept for student Leap Card application triage using OCR extraction, rule-based validation, CSV audit reports, and email notifications; targeted **≥50%** auto-approval and **60–70%** lower average handling time per case.

B.Tech, Electronics & Communication Engineering — SRIN College, India

Jun 2016 – Nov 2020

SKILLS & TECHNOLOGIES

Cloud Skills

Azure Data Factory, Azure Databricks, Microsoft Fabric, Lakehouse, OneLake, Azure SQL, Power BI Service and Desktop, Data modeling, Event Streams, Data Activator, KQL Database, Lakehouse Medallion architecture, Delta Lake, Spark SQL, Databricks Jobs, Fabric Data Warehouse, SQL Analytics Endpoint, Direct Lake, Fabric semantic models, Fabric Notebooks, Fabric Data Pipelines.

Business Intelligence | Data Engineering

T-SQL, MS SQL Server, SSMS, SSIS, SSRS Paginated Reports, Power BI (DAX, Power Query), Tidal Automation, Jaspersoft, RapidMiner, ADF Pipelines, Databricks Workflows, ETL/ELT pipelines and batch processing, PySpark transformations, notebook orchestration, configuration-driven ingestion, medallion layer design, Bronze/Silver/Gold processing, Delta tables, schema validation, data quality rules, audit logging, incremental processing, batch orchestration, source-to-target mapping, data reconciliation, row-count checks, aggregate checks, exception handling, report migration validation, pipeline monitoring, performance tuning.

Programming | ML

SQL, PySpark, Python, KQL (Kusto Query Language), NumPy, Pandas, Matplotlib, Scikit-learn, TensorFlow, Supervised, Unsupervised, Reinforcement Learning, Recommendation Systems, Feature Engineering, Model Evaluation, Feature Scaling, NLP, ANN, DNN, Keras, ML-ready datasets, predictive modelling, optimisation algorithms, RPA with UiPath.

Business Tools | DevOps

Jupyter Notebook, Google Colab, VS Code, Git, Databricks Workflows, ADF Pipelines, Tidal Automation, Jira, Confluence, Agile delivery (Scrum, Kanban), DEV/INT/CRT/UAT/EUAT/PRE-PROD/PROD environments, Data governance and risk compliance, TFS, CI/CD release controls.

CERTIFICATIONS

- Microsoft Certified: Azure Fundamentals (AZ-900) | Microsoft | Dec 2024
- Microsoft Certified: Azure AI Fundamentals (AI-900) | Microsoft | Dec 2024
- Microsoft Certified: Power BI Data Analyst (PL-300) | Microsoft | Dec 2024
- Microsoft Certified: Fabric Data Engineer Associate (DP-700) | Microsoft | Mar 2026
- Supervised Machine Learning: Regression and Classification | DeepLearning.AI | Dec 2025
- Advanced Learning Algorithms | DeepLearning.AI | Jan 2026
- Unsupervised Learning, Recommenders, Reinforcement Learning | DeepLearning.AI | Jan 2026
- Oracle Cloud Infrastructure AI Foundations Associate (1Z0-1122-25) | Oracle | Sep 2025
- Programming for Everybody (Getting Started with Python) | University of Michigan (Coursera) | Aug 2025.
- 20+ Professional Courses in SQL, SSIS, ADF, Databricks, Power BI, PySpark, Microsoft Fabric, Kafka | Udemy · DataCamp · Coursera · Microsoft Learn | Ongoing

AWARDS & RECOGNITION

- Best Employee of the Project - Caresource MDP migration | Cognizant | 2024.
- Emerging Talent Award | Cognizant | 2023.
- “Working as One” Award | Cognizant | 2022.